



OFFICIAL VISUAL COMPANION

Inside GeoBusiness Intelligence Studio

A Visual Guide to Open Data, Geospatial Discovery, and Local Business Intelligence

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PURPOSE

Why build a local-business intelligence laboratory?

Business-location data often arrives fragmented: names vary, addresses are incomplete, contact fields disappear, and the same place may appear more than once. GeoBusiness Intelligence Studio converts that ambiguity into an inspectable pipeline.

Accessible core without a paid AI key
Visible provenance for every provider
Conservative duplicate inference
Portfolio-ready engineering depth

PROVIDERS

One interface, three data paths

Offline fictional samples support predictable demonstrations. OpenStreetMap uses Nominatim and Overpass with safeguards. The optional Google Places adapter uses the official API and the user's own restricted key.

Offline fictional sample provider
OpenStreetMap via Nominatim and Overpass
Optional official Google Places API
Replaceable adapter boundary

WORKFLOW

From a query to an exportable dataset

Choose provider, business type, city, radius, and limit. Resolve geography, retrieve records, normalize fields, score completeness, remove probable duplicates, map results, summarize coverage, and export CSV or JSON.

Choose provider and geography
Resolve and retrieve records
Normalize and score quality
Map, inspect, analyze, and export

INTERFACE

A dashboard that keeps evidence visible

Search controls expose provider and geography. Metric cards reveal coverage and average quality. The map preserves spatial context, while tables keep source, contacts, ratings, and quality visible.

Search controls expose assumptions
Metric cards summarize coverage
Map preserves spatial context
Table and exports retain provenance

ARCHITECTURE

FastAPI at the center of replaceable components

Provider adapters feed validated FastAPI routes. Search orchestration performs normalization, quality scoring, sorting, and conservative deduplication. Static HTML and Leaflet provide the browser interface.

Provider adapters
Validated API routes
Search orchestration
Map, table, analysis, CSV, JSON

DATA QUALITY

Completeness is measured; identity is inferred cautiously

The quality score measures the presence of useful fields, not factual correctness.

Duplicate detection combines stronger evidence such as matching phone numbers with softer evidence such as similar names and addresses.

Completeness is not correctness
Phone equality is strong evidence
Name and address similarity are softer
Users can disable deduplication

PRIVACY · AI · DEPLOYMENT

Useful without AI; extensible when justified

The deterministic workflow is complete without AI. Ollama and OpenAI-compatible endpoints are optional. The project supports Render, Docker, source execution, and a verified one-file Windows edition.

Render browser application
Verified Windows EXE
Python and Docker modes
Nominatim and Overpass safeguards

USE CASES · CITATION

Research value with permanent distribution

Use the project for market orientation, data-engineering education, responsible-AI demonstrations, and portfolio review. Version 1.2.0 is preserved on Zenodo and identified by a permanent DOI.

Market orientation
Data-engineering education
Responsible-AI demonstrations
Zenodo DOI and versioned preservation

Windows: <https://github.com/FaramarzKowsari/geo-business-intelligence-studio/releases/latest/download/GeoBusiness-Intelligence-Studio.exe>

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The map becomes intelligence only after the uncertainty remains visible.